

Introduction

```

[Temperature Sensor] → ADC →
                        → STM32 → PWM → Fan
[MPU6050 (I2C)] → I2C →      → GPIO → LED
                  → SPI → OLED Display
                  → UART → PC
[Button] → Interrupt →

```

Hardware Components

- STM32F407VG
- Temperature sensor (e.g., LM35)
- MPU6050 (I2C)
- OLED Display (SPI)
- DC Fan / Motor
- Push Button
- LED

Software Design

6.1 GPIO

- LED ON/OFF control
- Button input

6.2 Interrupt

- Button press changes system mode

6.3 ADC

- Reads temperature sensor

6.4 PWM

- Controls fan speed based on temperature

6.5 I2C

- Reads motion data from MPU6050

6.6 SPI

- Sends data to OLED display

6.7 UART

- Sends debug data to PC

6.8 Timer

- Executes periodic reading every 1 second

Control Algorithm

```
if (temperature > threshold)
{
    increase_PWM();
}
```

```
if (motion_detected)
{
    LED_ON();
}
```

SPI Communication Explanation

SPI is:

- Full-duplex
- High-speed
- Uses MOSI, MISO, SCK, CS lines

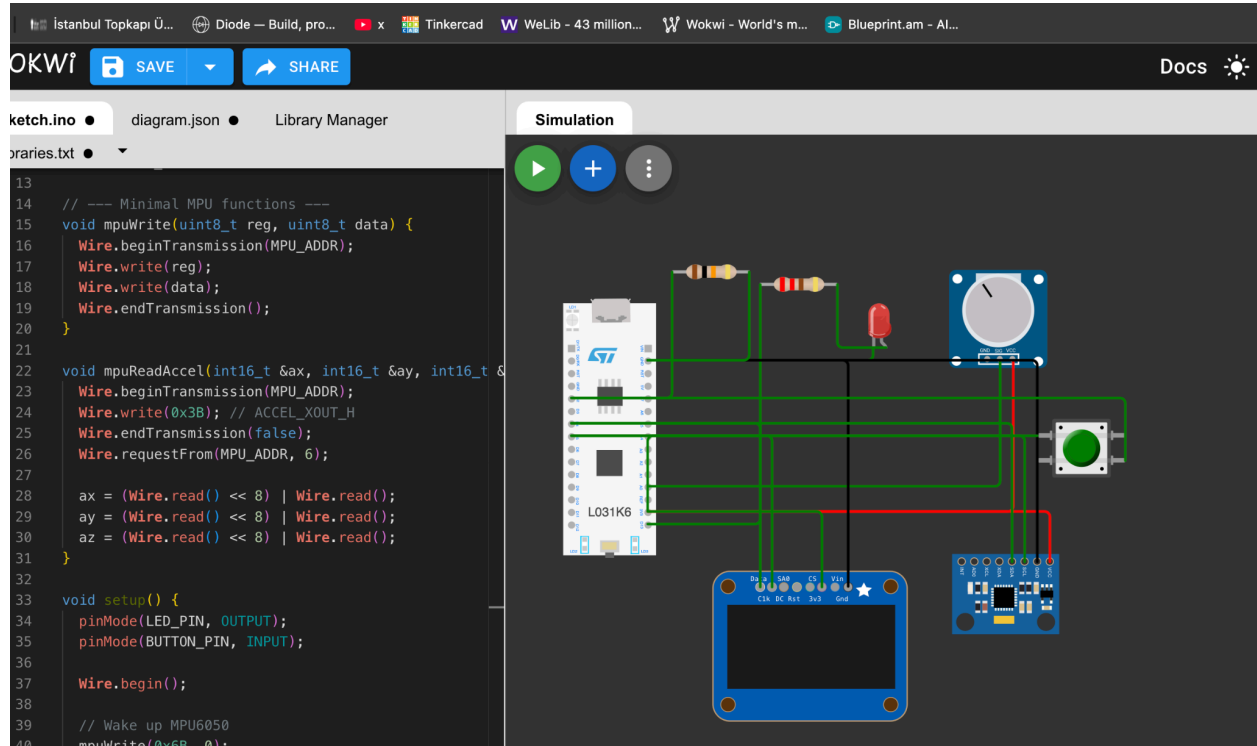
Used here to:

➡ Send display data to OLED

Expected Results

- Temperature increases → Fan speed increases
- Motion detected → LED turns ON
- Data displayed on OLED
- Data sent to PC via UART

Simulation(Wokwi)



Conclusion

The system successfully integrates multiple peripherals and communication protocols. SPI proves efficient for display communication, while ADC, I2C, and PWM enable real-time sensing and control.